

Regional Income Growth and House Price Dynamics in the Netherlands

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Set-up your environment

Regional Income Growth and House Price Dynamics in the Netherlands

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Tutorial group 3.3

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

The Dutch housing market has been under pressure for several years now. Between 2013 and 2023 the average price of houses has more than doubled, while the income growth in the same period considerably lagged behind. The increasing gap between housing costs and purchasing power is not only an economic phenomenon, but also a social problem; it determines who does have access to a stable living situation and who doesn't.

It's striking that this development differs significantly across regions in the Netherlands. In the Randstad, prices increased much more than in the peripheral regions like Groningen or Zeeland, while the income development per region differs significantly too. Factors such as access to big employers, education level and local economic structure play an important role in this shift. And it is because of this regional variety that it's interesting to investigate the relation between income and house prices.

This project investigates the relationship between regional income growth and house price dynamics in Dutch provinces, based on CBS-data and the programming tool R. The goal is to discover in which regions the housing market detached most from the local income development.

Part 2 - Data Sourcing

2.1 Load in the data

we used 9 different data sets and combined we did this using this code for each first we cleaned the already existing datasets. and combined everything together using this code: `datacombined15tot23 <-`

```
rbind(Data15161718, data2019, data2020, data2021, data2022, data2023).
```

```
#Bev_dichtheid <- read.csv("C:/Users/vince/Downloads/Regionale_kerncijfers_Nederland_10062026_163720.csv")

Bev_dichtheid <- read.csv("data/regionale_kerncijfers.csv", header=FALSE, sep=";")
write.csv(Bev_dichtheid, "goede data/bevolkingsdichtheid.csv")
```

joining

2.2 Provide a short summary of the dataset(s)

For this study three datasets from the Centraal Bureau voor de Statistiek (CBS) were used. The first dataset, Bestaande koopwoningen; verkoopprijzen prijsindex 2020=100 (85773NED), contains data on the variables ‘gemiddelde verkoopprijs van bestaande koopwoningen en bijbehorende prijsindex op provinciaal niveau’. The second dataset, inkomen van huishoudens; huishoudenskenmerken, regio (86004NED), contains information on the ‘gemiddelde inkomen van huishoudens, op provinciaal niveau’. Lastly, the third dataset, bevolkingsdichtheid op 1 januari(70072NED), contains information on the population density in the Netherlands.

All datasets are available on CBS StatLine. The dataset on the prices of houses spans from 2015 until 2025, the dataset on income only covers 2015 until 2023 (In order to see other years or regions, the filter option should be used to select the desired years). The dataset on population density in the Netherlands spans from 1995 until 2025. For the analysis in this research the overlapping years of 2015-2023(or a part thereof) are studied.

2.3 Describe the type of variables included

The datasets mostly have social economic variables (SES). The house price dataset measures the average sale price of existing houses and the price index of these sales. This is an indicator of the accessibility of the housing market in the Netherlands. The dataset on the income of households measures the average income of households per province, as a benchmark for wealth in that region. The dataset on population density, again, is based on administrative data. It measures the amount of inhabitants per square kilometer in every province.

The data originates from administrative sources, not interviews or questionnaires. The average house prices are based on the ‘Basisregistratie Kadaster’, in which every official sale of a house is recorded by law. The data on the income of households are deduced from tax returns and benefit registrations, processed by the CBS into regional stats. The data on population density is computed by dividing the amount of inhabitants officially listed by the amount of square kilometers in the region.

Administrative data mostly doesn’t suffer from non-responses and measurement errors. A possible challenge is the fact that administrative definitions change over time; income can be defined differently from one year onto the next. This could affect the comparability over the years.

Part 3 - Quantifying

3.1 Data cleaning

Data cleaning datacombined15tot23goed. we mainly used this code to clean this data set: `names(datacombined15tot23goed)[names(datacombined15tot23goed) == "X1.000.euro...6"] <- "gem_st_inkomen_x_100"` ETC.

Data cleaning woningprijsprov: we mainly used this code to clean this data set: `woningprijsprov‘1995’ <- format(woningprijsprov1995, big.mark = ",", scientific = FALSE)`

```
library(tidyverse)
bevolkingsdichtheid <- Bev_dichtheid
bevolkingsdichtheid <- bevolkingsdichtheid %>% filter(V1 != "Regionale kerncijfers Nederland")
bevolkingsdichtheid <- bevolkingsdichtheid %>% filter(V1 != "Perioden")
bevolkingsdichtheid <- bevolkingsdichtheid %>% filter(V1 != "Onderwerp")
bevolkingsdichtheid <- bevolkingsdichtheid %>% filter(V1 != "2024")
bevolkingsdichtheid <- bevolkingsdichtheid %>% filter(V1 != "Bron: CBS")
bevolkingsdichtheid$...1 <- NULL

write.csv(bevolkingsdichtheid, "goede data/goede_bevolkingsdichtheid.csv", row.names = FALSE)
```

```
library(tidyverse)

# Het CBS-bestand is 'breed' en heeft 6 koprijen: titel, 2 lege regels,
# 'Onderwerp', de provincienamen (rij 5) en de eenheid (rij 6). De data
# beginnen pas op rij 7. Daarom lezen we zonder header en bouwen we zelf
# een lang bestand met de kolommen Perioden / Regio.s / dichtheid.
ruw <- read.csv(
  "data/regionale_kerncijfers.csv",
  sep = ";",
  header = FALSE,
  strip.white = TRUE,
  fileEncoding = "UTF-8-BOM"
)

provincienamen <- as.character(unlist(ruw[5, -1]))
bevolking <- ruw[7:nrow(ruw), ]
colnames(bevolking) <- c("Perioden", provincienamen)

bevolking <- bevolking %>%
  pivot_longer(
    cols = -Perioden,
    names_to = "Regio.s",
    values_to = "bevolkingsdichtheid_km2"
  )

bevolking_breed <- bevolking %>%
  filter(Regio.s != "Nederland") %>%
  mutate(
    Regio.s = gsub(" \\(PV\\)", "", Regio.s),
    bevolkingsdichtheid_km2 = as.numeric(bevolkingsdichtheid_km2)
  ) %>%
  pivot_wider(
    names_from = Perioden,
    values_from = bevolkingsdichtheid_km2
  ) %>%
  arrange(Regio.s)

head(bevolking_breed) # View() werkt niet tijdens het knitten
```

```
## # A tibble: 6 x 6
##   Regio.s '2021' '2022' '2023' '2024' 'Bron: CBS'
##   <chr>      <dbl> <dbl> <dbl> <dbl>      <dbl>
```

## 1	1071	1084	1093	1109	1118	NA
## 2	1361	1380	1391	1410	1423	NA
## 3	187	188	189	191	191	NA
## 4	194	195	196	197	198	NA
## 5	215	216	217	220	220	NA
## 6	251	253	255	257	260	NA

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1 creating new variable 1, we have created the average disposable income of each province over 8 years. using the data set: `gem_inkomen_provincie`, We used the following code for this: `gem_inkomen_provincie <- aggregate(gem_bstb_inkomen_x_100 ~ Provinces, data = datacombined15tot23goed, FUN = mean)`

Variable 2 creating new variable 2 using the original dataset: `datacombined15tot23goed`, in which we represented the weighted averages of income for different provinces: inside and outside the Randstad.

adjust the format of the current cleaned dataset in order to continue working with a renewed dataset, and also a renewed dataset, because for this new variable we actually need to merge our two datasets.

```
woningprijsprov <- read.csv("goede data/goedehuizenprijzenperprov.csv", check.names = FALSE)

woningprijsprov <- woningprijsprov[!grepl("Bron", woningprijsprov$`Regio.s`), ]

woningprijsprov$Onderwerp <- NULL
woningprijsprov$`...4` <- NULL

woningprijsprov_lang <- reshape(woningprijsprov,
                                varying = c("2015", "2020", "2021", "2022", "2023"),
                                v.names = "gem_huizenprijs",
                                timevar = "jaar",
                                times = c(2015, 2020, 2021, 2022, 2023),
                                direction = "long")

woningprijsprov_lang$id <- NULL

write.csv(woningprijsprov_lang, "goede data/goedehuizenprijzenperprov_lang.csv", row.names = FALSE)
```

variabele 3:

```
bevolkingsdichtheid_raw <- read.csv("goede data/goede_bevolkingsdichtheid.csv", header = FALSE)

bevolkingsdichtheid_raw <- bevolkingsdichtheid_raw[2:6, 1:14]

colnames(bevolkingsdichtheid_raw) <- c("jaar", "Nederland", "Groningen (PV)", "Fryslân (PV)",
                                         "Drenthe (PV)", "Overijssel (PV)", "Flevoland (PV)",
                                         "Gelderland (PV)", "Utrecht (PV)", "Noord-Holland (PV)",
                                         "Zuid-Holland (PV)", "Zeeland (PV)", "Noord-Brabant (PV)",
                                         "Limburg (PV)")
```

```

bevolkingsdichtheid_raw <- bevolkingsdichtheid_raw[, !colnames(bevolkingsdichtheid_raw) %in% "Nederland"]

bevolkingsdichtheid_lang <- reshape(bevolkingsdichtheid_raw,
                                     varying = colnames(bevolkingsdichtheid_raw)[-1],
                                     v.names = "bevolkingsdichtheid_per_provincie",
                                     timevar = "Provincies",
                                     times = colnames(bevolkingsdichtheid_raw)[-1],
                                     direction = "long")

bevolkingsdichtheid_lang$id <- NULL
bevolkingsdichtheid_lang$jaar <- as.integer(bevolkingsdichtheid_lang$jaar)

## Warning: NAs introduced by coercion

bevolkingsdichtheid_lang$bevolkingsdichtheid_per_provincie <- as.integer(bevolkingsdichtheid_lang$bevolkingsdichtheid_per_provincie)

## Warning: NAs introduced by coercion

bevolkingsdichtheid_lang <- bevolkingsdichtheid_lang[bevolkingsdichtheid_lang$jaar %in% c(2020, 2021, 2022, 2023), ]

huizenprijzen_raw <- read.csv("goede data/goedehuizenprijzenperprov.csv")
huizenprijzen_raw <- huizenprijzen_raw[!grepl("Bron", huizenprijzen_raw$Regio.s), ]

huizenprijzen_schoon <- data.frame(
  Provincies = huizenprijzen_raw$Regio.s,
  X2020 = huizenprijzen_raw$X2020,
  X2021 = huizenprijzen_raw$X2021,
  X2022 = huizenprijzen_raw$X2022,
  X2023 = huizenprijzen_raw$X2023
)

huizenprijzen_lang <- reshape(huizenprijzen_schoon,
                              varying = c("X2020", "X2021", "X2022", "X2023"),
                              v.names = "gem_huizenprijs",
                              timevar = "jaar",
                              times = c(2020, 2021, 2022, 2023),
                              direction = "long")

huizenprijzen_lang$id <- NULL

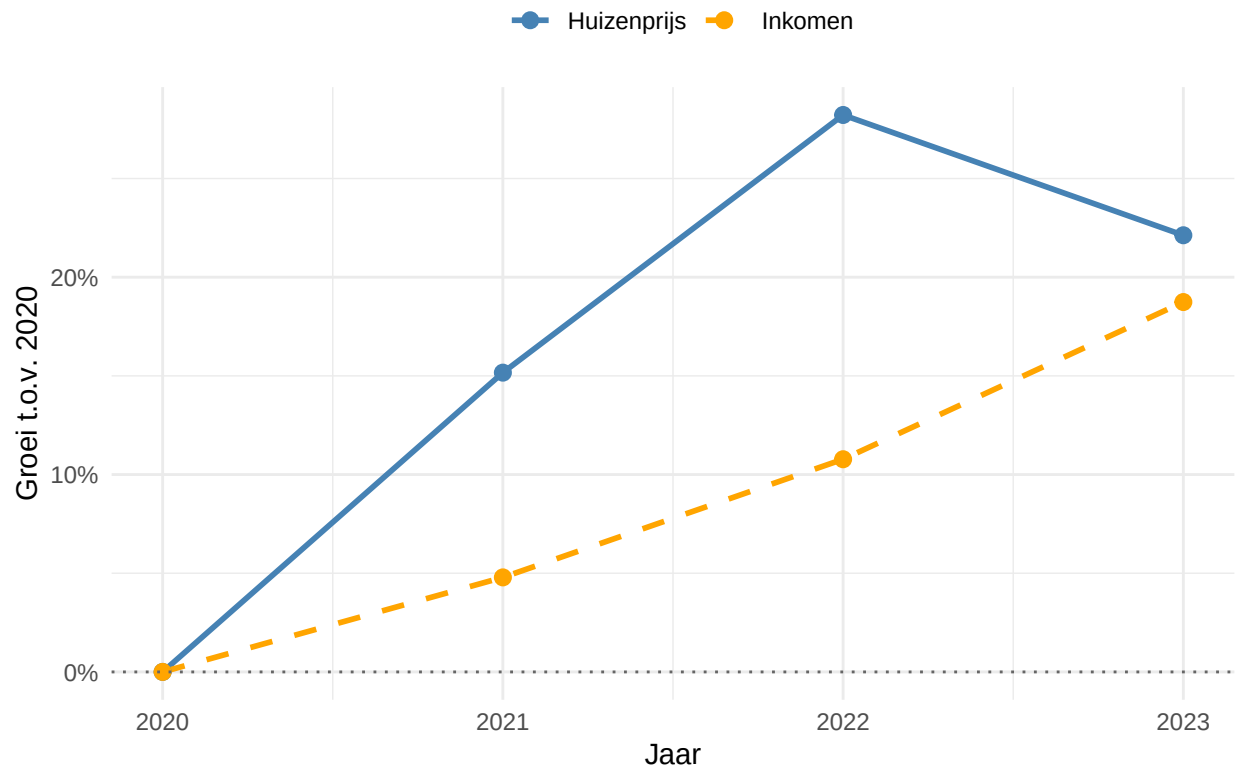
bevolking_huizen <- merge(bevolkingsdichtheid_lang, huizenprijzen_lang[, c("Provincies", "jaar", "gem_huizenprijs")],
                          by = "Provincies")

write.csv(bevolking_huizen, "goede data/bevolkingsdichtheid_en_huizenprijzen_per_provincie.csv", row.names = FALSE)

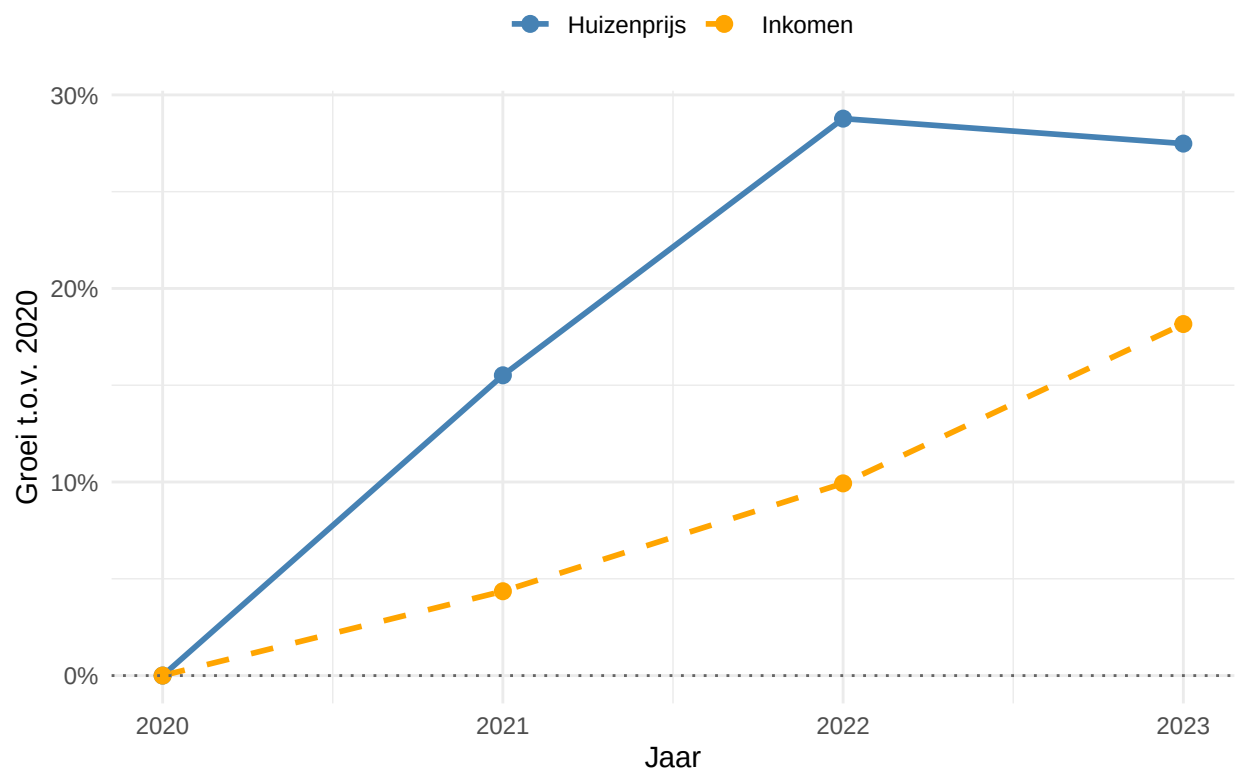
```

3.3 Visualize temporal variation

Groei Randstad t.o.v. 2020



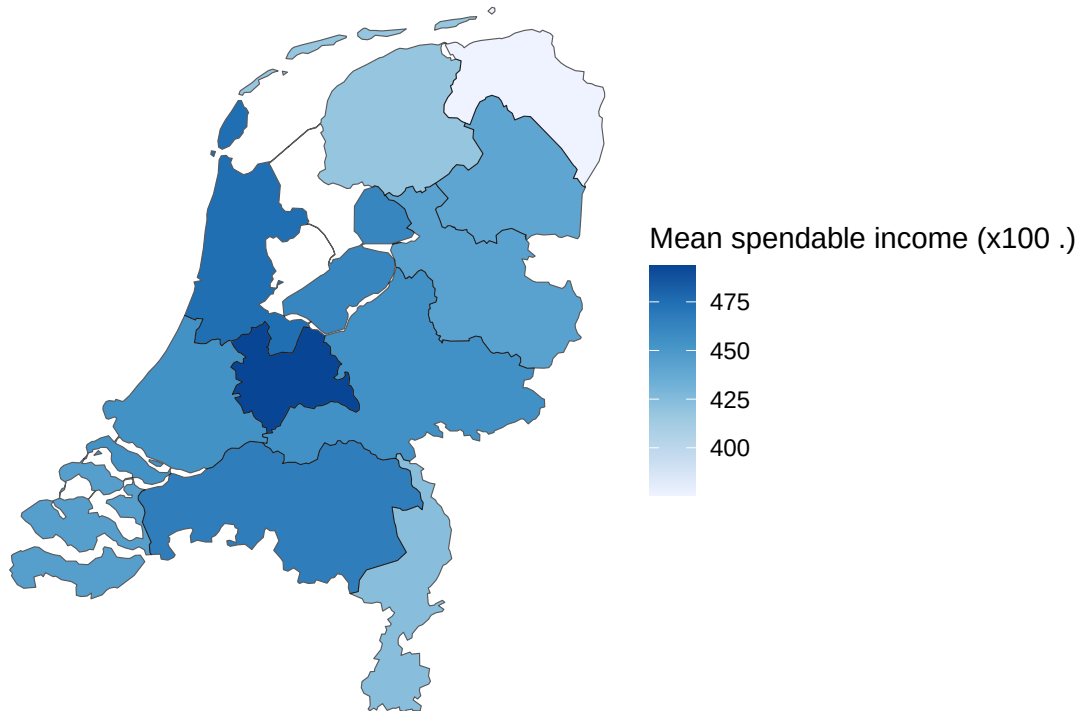
Groei Buiten Randstad t.o.v. 2020



3.4 Visualize spatial variation

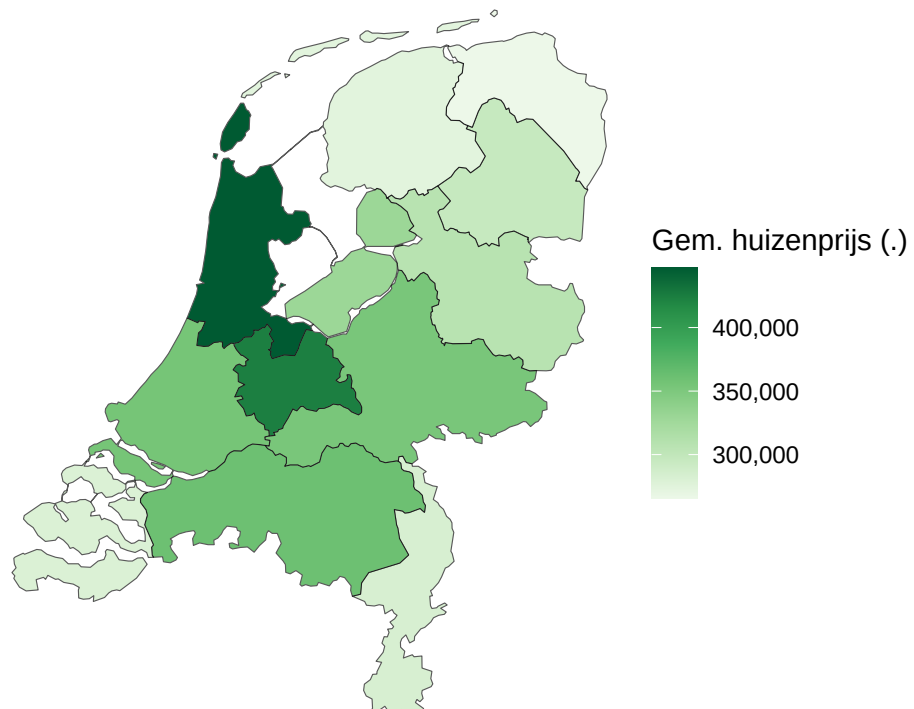
Mean Spendable Income per Province

Average over 2015–2023



Gemiddelde Huizenprijs per Provincie

Gemiddelde over de laatste jaren



Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

What is the poverty rate by state?

```
library(tidyverse)

df_provincies <- read.csv("goede data/bevolkingsdichtheid_en_huizenprijzen_per_provincie.csv", stringsAsFactors = FALSE)

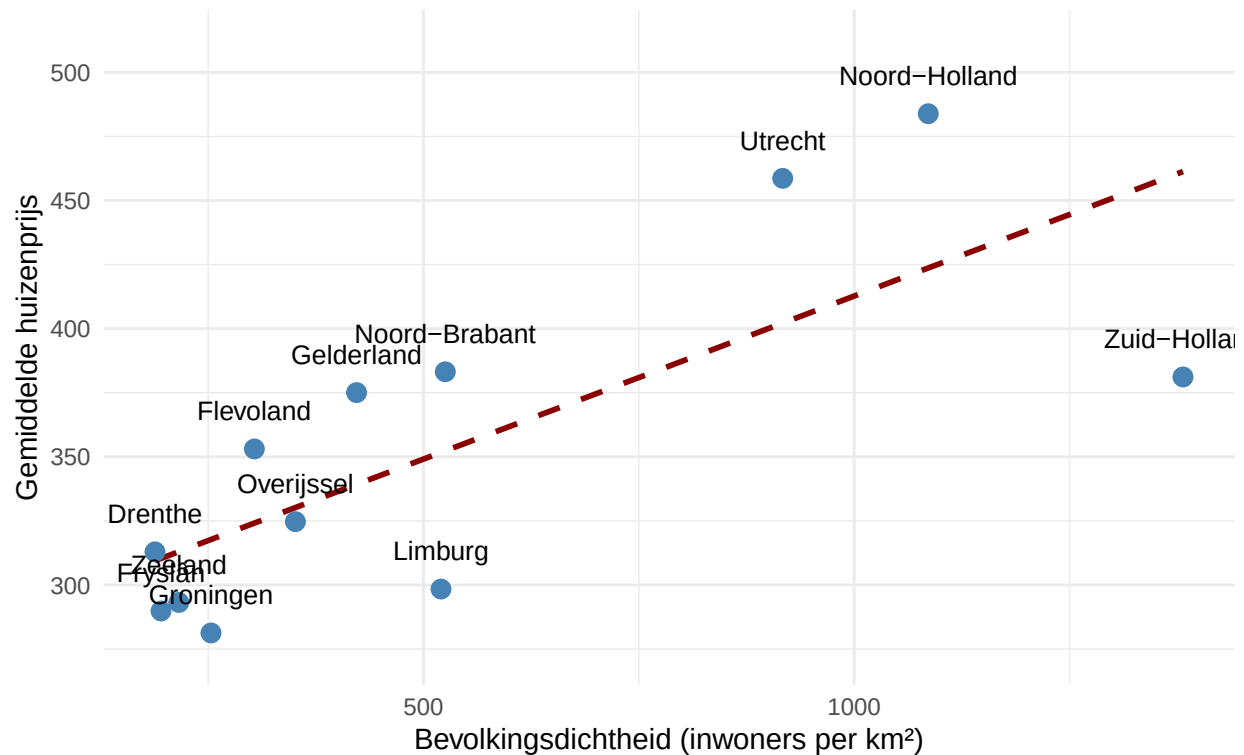
df_plot <- df_provincies %>%
  mutate(Provincies = str_remove(Provincies, " \\(PV\\)")) %>%
  group_by(Provincies) %>%
  summarise(
    dichtheid = mean(bevolkingsdichtheid_per_provincie, na.rm = TRUE),
    huizenprijs = mean(gem_huizenprijs, na.rm = TRUE)
  ) %>%
  drop_na(dichtheid, huizenprijs)

ggplot(data = df_plot, aes(x = dichtheid, y = huizenprijs, label = Provincies)) +
  geom_smooth(method = "lm", formula = y ~ x, color = "darkred", linetype = "dashed", se = FALSE) +
  geom_point(color = "steelblue", size = 3) +
  geom_text(vjust = -1.5, size = 3.5) +
  scale_y_continuous(expand = expansion(mult = c(0.1, 0.2))) +
  labs(
    title = "Correlatie Bevolkingsdichtheid vs. Huizenprijzen (2020-2022)",
    subtitle = "Met lineaire regressielijn",
    x = "Bevolkingsdichtheid (inwoners per km2)",
    y = "Gemiddelde huizenprijs"
  ) +
  theme_minimal()
```

```
## Warning: The following aesthetics were dropped during statistical transformation: label.
## i This can happen when ggplot fails to infer the correct grouping structure in
##   the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
##   variable into a factor?
```


Correlatie Bevolkingsdichtheid vs. Huizenprijzen (2020–2022)

Met lineaire regressielijn



Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Analyze the relationship between two variables.

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the to your PUBLIC repository here: ...

5.2 Reference list

Centraal Bureau voor de Statistiek (2024). Inkomens van huishoudens; huishoudenskenmerken, regio [Dataset]. CBS. <https://opendata.cbs.nl/#/CBS/nl/dataset/86004NED/table>

Centraal Bureau voor de Statistiek. (2026). Bestaande koopwoningen; verkoopprijzen prijsindex 2020=100 [Dataset]. CBS. <https://opendata.cbs.nl/#/CBS/nl/dataset/85773NED/table>

Centraal Bureau voor de Statistiek. (2026). Regionale kerncijfers Nederland [Dataset].CBS. <https://opendata.cbs.nl/#/CBS/nl/dataset/70072ned/table?dl=17F53>